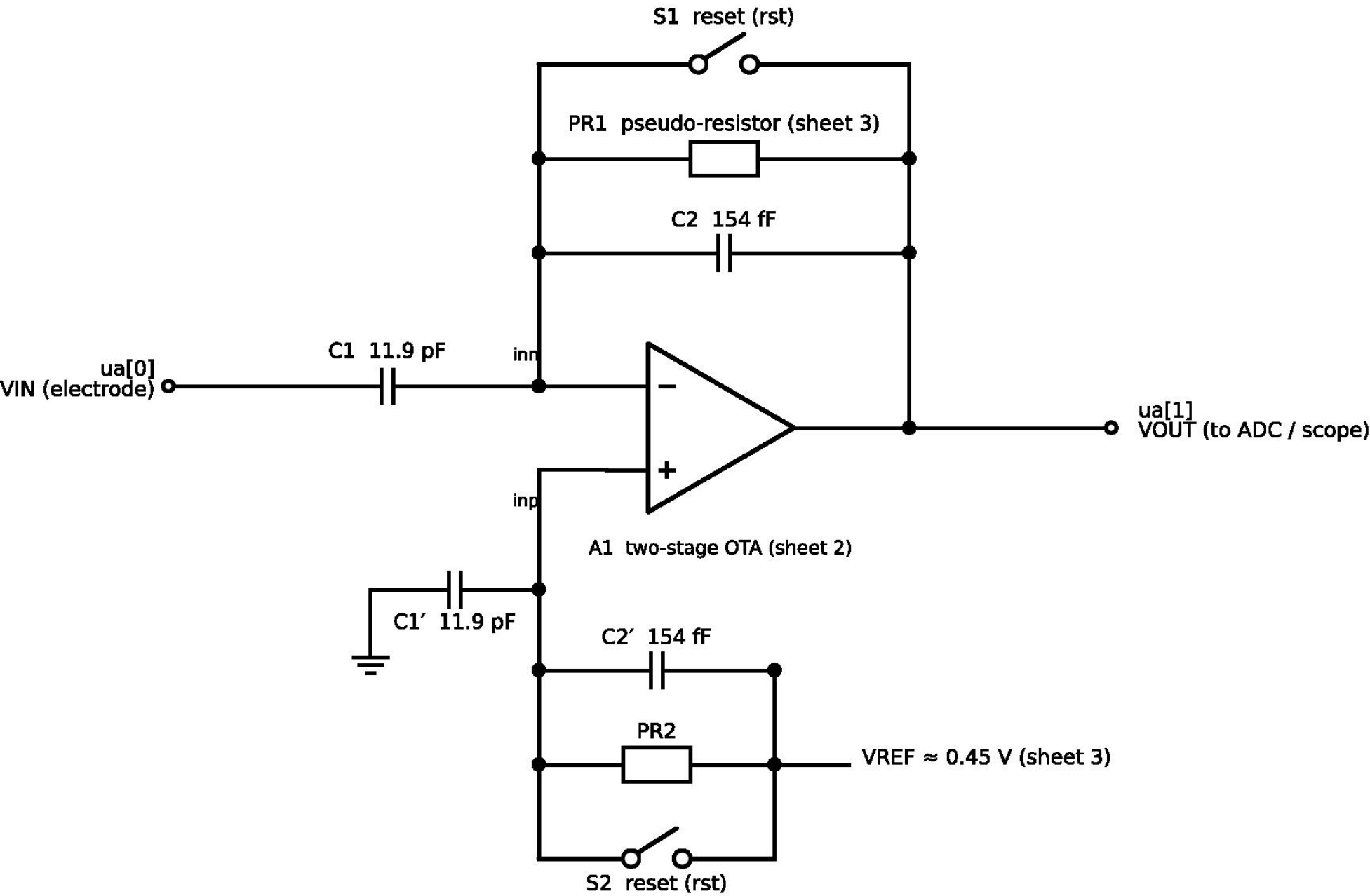
Sheet 1 of 4 — Neonatal EEG front-end (top level, simulated design v1)

Capacitive-feedback amplifier · SKY130 · Tiny Tapeout 1x2 tile · ua[0] in, ua[1] out, ui_in[0] = rst



Midband gain = C1/C2 = 77.5 V/V (37.8 dB). High-pass $f_L = 1/(2\pi \cdot R_{PR} \cdot C2) \approx 0.09$ Hz; low-pass ≈ 22 kHz (set by OTA bandwidth).

C1′ goes to ground: VREF noise reaches VOUT with gain ≈ 1, not 78, so a simple on-chip VREF is good enough.

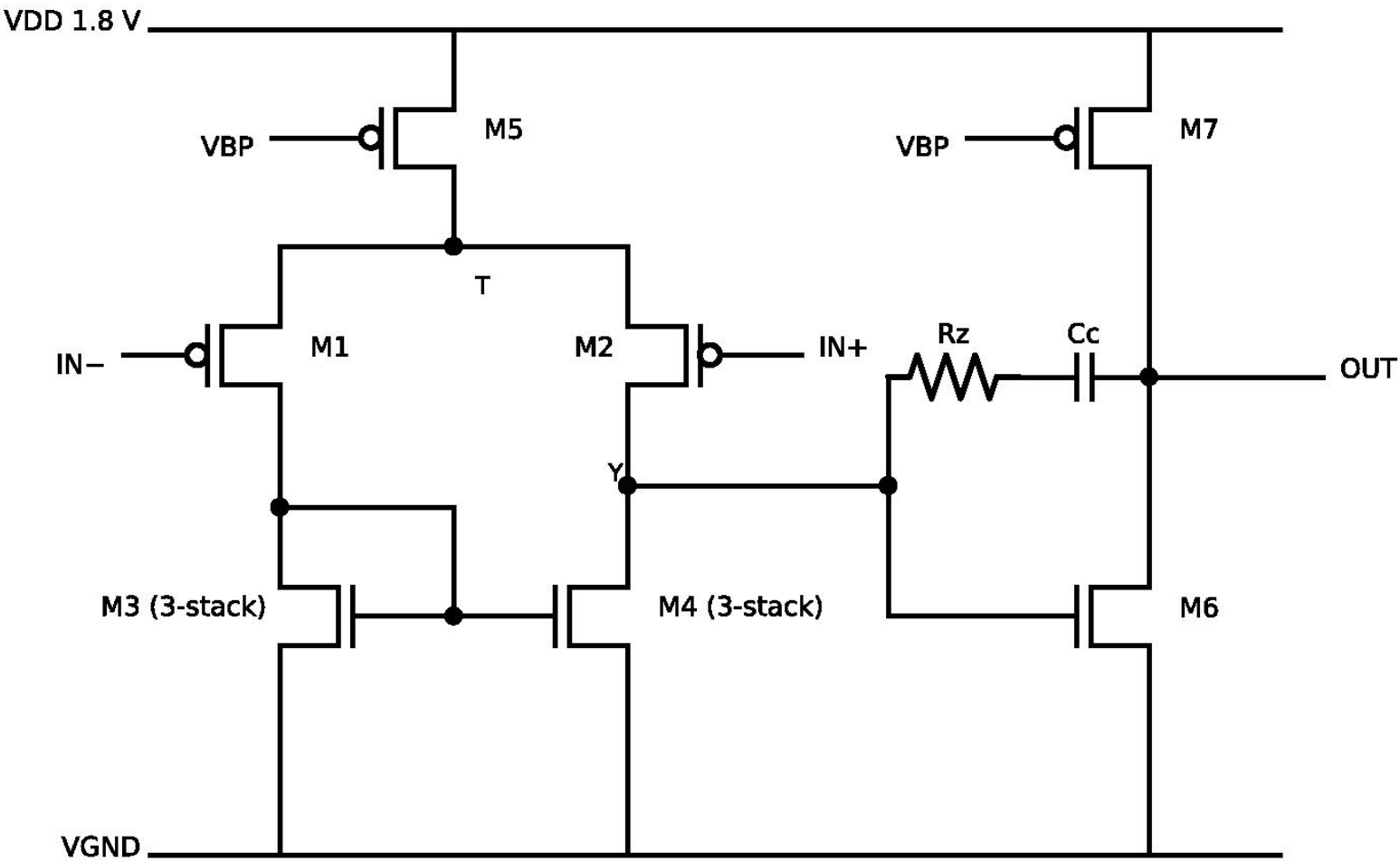
S1/S2 are minimum NMOS switches (0.42/0.15 µm) driven by rst = ui_in[0]. Holding rst high forces VOUT = VREF:

use it for ~10 ms at power-up and after electrode artifacts (pseudo-resistor recovery alone takes seconds).

MIM capacitors: C1/C1′ = cap_mim_m3_1 77×77 µm, C2/C2′ = 8.6×8.6 µm. Unused digital outputs are tied to VGND (not drawn).

Sheet 2 of 4 — A1: two-stage Miller OTA

PMOS input pair, NMOS mirror load (3-device stacks for low 1/f noise), NMOS common-source 2nd stage, Rz–Cc compensation



IN− = gate of M1 (inverting), IN+ = gate of M2 (non-inverting).

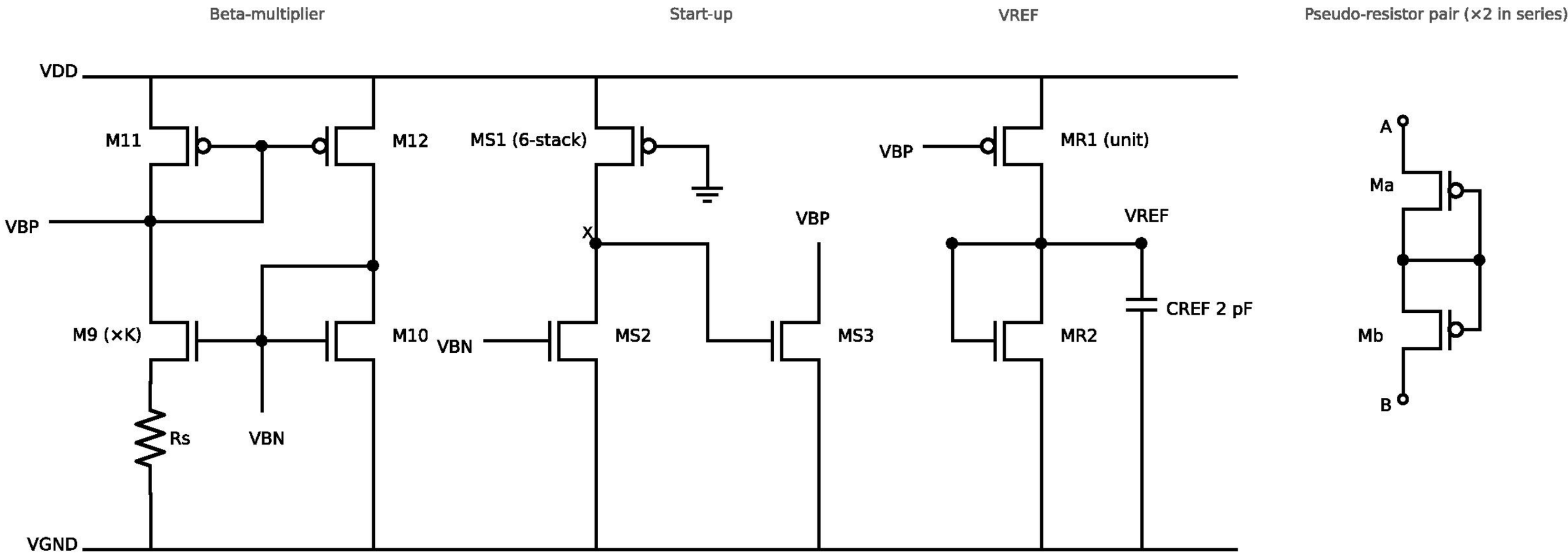
Bulks: M1/M2 tied to tail node T (own n-well); other PMOS to VDD; all NMOS to VGND.

M3 and M4 are each three 7/20 µm NMOS in series (effective L = 60 µm); long L lowers the mirror's input-referred 1/f noise.

VBP comes from the bias generator on sheet 3 (0.5 µA per PMOS unit cell 4/2 µm). Sizes on sheet 4.

Sheet 3 of 4 — Bias generator, start-up, VREF and pseudo-resistor

Subthreshold beta-multiplier: I_REF ≈ n·U_T·ln(K) / Rs = 0.50 µA at 27 °C (0.43–0.62 µA over corners) · K = 4, Rs = 92 kΩ



Start-up: if no current flows, VBN = 0 → MS2 off → MS1 pulls X high → MS3 pulls VBP down and starts the loop.

Once running, VBN rises → MS2 pulls X low → MS3 turns off.

MS1 is six 0.6/20 µm PMOS in series (weak pull-up). Simulated: X < 20 mV once started, all corners, both model sets.

VREF = Vgs of a weak-inversion NMOS diode at 0.5 µA → 0.45 V typical (0.39–0.50 V over corners, 0–70 °C).

Pseudo-resistor = two Ma/Mb pairs in series (≈1.3×10^{13} Ω at 27 °C). Each PMOS 0.42/1 µm, own n-well, bulk to outer node.

Reset switches (0.42/0.15 µm NMOS, gate = rst) tie the internal nodes to A so the chain restarts at zero bias.

Sheet 4 of 4 — Device sizes (SKY130 1.8 V devices, W / L in µm)

Simulated in ngspice with SKY130 models: 5 process corners × 0/27/37 °C × 1.71/1.8/1.89 V, classic and continuous models; specified 0–50 °C. Pre-layout.

Device	SKY130 primitive	Size	Role
Front-end (sheet 1)			
C1, C1′	cap_mim_m3_1	77 × 77 µm (11.9 pF)	Input coupling
C2, C2′	cap_mim_m3_1	8.6 × 8.6 µm (154 fF)	Feedback; gain = C1/C2 = 77.5 V/V
PR1, PR2	pfet_01v8 ×4 each	0.42 / 1	Two pairs in series, ≈1.3×10¹³ Ω
S1, S2 + 3 per PR	nfet_01v8	0.42 / 0.15	Reset switches, gate = rst (ui_in[0])
A1 OTA (sheet 2)			
M1, M2	pfet_01v8	5 / 8 × 30 fingers	2 µA each; W = 150 µm total
M3, M4	nfet_01v8	7 / 20, 3 in series	Mirror load, effective L = 60 µm
M5	pfet_01v8	4 / 2 × 8 units	4 µA tail
M6	nfet_01v8	5 / 2 × 4	10 µA second stage
M7	pfet_01v8	4 / 2 × 20 units	10 µA load
Cc, Rz	MIM / xhigh_po	2 pF, 20 kΩ	Miller compensation (ideal in sim)
Bias, start-up, VREF (sheet 3)			
M11, M12, MR1	pfet_01v8	4 / 2	Unit cell, 0.5 µA
M10 / M9	nfet_01v8	7 / 1 × 2 / × 8	K = 4
Rs	res_xhigh_po_0p35	L = 16 µm (92 / 118 kΩ)*	Sets I_REF
MS1 / MS2 / MS3	pfet / nfet / nfet	6×0.6/20 · 7/1×2 · 0.42/2	Start-up
MR2	nfet_01v8	7 / 1 × 6	VREF diode → 0.45 V
CREF	cap_mim_m3_1	2 pF	VREF decoupling (ideal in sim)

Supply current 15.7 µA typical (13–17 µA over corners, 0–37 °C) → about 28 µW at 1.8 V. *Rs = 92 kΩ in the classic SKY130 models, 118 kΩ in the continuous (combined) models → 0.50 / 0.39 µA unit current, 15.7 / 12.2 µA supply.

MIM total ≈ 28 pF ≈ 14,000 µm², about 40% of the 160 × 225 µm 1×2 tile. MIM uses met3/met4: keep it clear of Tiny Tapeout's met4 power stripes; met5 is not allowed.

All transistors stay inside the SKY130 model bins (finger W ≤ 7 µm, L ≤ 20 µm); wide devices use parallel fingers.

Next: enter in Xschem, re-simulate, then layout (Magic/KLayout), DRC, LVS, extraction and post-layout simulation.